PassFrame
tuition

Perimeter & Area

Year 5 Accelerated — 11+ Lesson Worksheet (PassFrame Tuition)

Worked examples — tutor talks through these first

Easy: A rectangle is 8 cm long and 5 cm wide. What is its perimeter?
Method & answer: Perimeter = 2 × (8 + 5) = 2 × 13 = 26 cm.

Standard: A rectangle is 12 cm long and 7 cm wide. What is its area?
Method & answer: Area = length × width = 12 × 7 = 84 cm².

Advanced: An L-shaped room is a 10m × 6m rectangle with a 4m × 3m rectangle removed from one corner. What is the area of the room?
Method & answer: Big rectangle = 10 × 6 = 60 m². Removed piece = 4 × 3 = 12 m². Area = 60 − 12 = 48 m².

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangle is 6 cm by 4 cm. What is its perimeter?

2. A square has sides of 5 cm. What is its perimeter?

3. Which of these has an area of 27cm²? A) a 9cm×3cm rectangle B) a 6cm×4cm rectangle C) a 5cm×5cm square D) a 8cm×3cm rectangle

4. A square has sides of 7 cm. What is its area?

5. A rectangle is 10 cm long and 4 cm wide. What is its perimeter?

6. A square has a perimeter of 24 cm. What is the length of one side?

7. A rectangle is 8 cm by 5 cm. What is its area?

8. A square has an area of 36 cm². What is the length of one side?

9. Select ALL of these shapes that have a perimeter of exactly 30cm: A) a 10cm×5cm rectangle B) a 12cm×3cm rectangle C) a square of side 7.5cm D) a 9cm×6cm rectangle E) an 11cm×4cm rectangle

10. A rectangle has a length of 12 cm and a width of 6 cm. What is its perimeter?

11. A square has sides of 9 cm. What is its area?

12. A rectangle is 15 cm long and 2 cm wide. What is its area?

13. Which is greater: the perimeter of an 8cm×6cm rectangle, or the perimeter of a 7cm square? A) the rectangle B) the square C) they are equal D) cannot tell

14. A square has sides of 11 cm. What is its perimeter?

15. A rectangle has an area of 40 cm² and a width of 5 cm. What is its length?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A triangle has a base of 10 cm and a height of 6 cm. What is its area?

2. A parallelogram has a base of 9 cm and a height of 4 cm. What is its

6. A triangle has an area of 24 cm² and a base of 8 cm. What is its height? A) 3cm B) 4cm C) 6cm D) 8cm

7. A rectangle is 1.5 m long and 80 cm wide. What is its area in cm²?

11. A rectangle's length is twice its width. If the width is 6 cm, what is the perimeter?

12. A triangle has a base of 14 cm and a height of 3 cm. What is its

area?

3. Which triangle has an area of 24cm²? A) base 8cm, height 6cm B) base 12cm, height 4cm C) both A and B D) neither

4. An L-shape is a 6cm×4cm rectangle with a 2cm×2cm square cut from one corner. What is its area?

5. A rectangular field is 40 m by 25 m. What is its perimeter?

8. Select ALL of these that have an area of MORE than 30cm²: A) a triangle base 10cm height 7cm B) a parallelogram base 8cm height 4cm C) a triangle base 12cm height 6cm D) a 5cm×6cm rectangle E) a triangle base 9cm height 6cm

9. Two rectangles, 5cm×3cm and 4cm×2cm, are joined with no overlap. What is the total area?

10. A square lawn has a perimeter of 48 m. What is its area? A) 144m² B) 121m² C) 96m² D) 169m²

area?

13. A rectangle has a perimeter of 50 cm and a length of 15 cm. What is its area?

14. An L-shaped garden is a 12m×8m rectangle with a 5m×3m rectangle removed from a corner. What is its area?

15. A 20cm×15cm photo is placed in a frame with a 2 cm border all around. What is the total area of the framed photo?

Standard / Hard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. A rectangular field is 60 m by 35 m. A path 2 m wide is built all the way around the inside edge. What is the area of the path?

2. A rectangular lawn is 18 m by 12 m. A 5m×3m flower bed is cut from one corner. What is the remaining area of the lawn?

3. Which of these compound shapes has the LARGEST area? A) an 8×5 rectangle minus a 2×2 corner B) a 6×6 square plus a triangle base 6 height 4 C) a triangle base 10 height 10 D) a 7×7 square

4. A triangle and a rectangle have equal areas. The rectangle is 8 cm by 5 cm, and the triangle has a base of 10 cm. What is the triangle's height?

5. A rectangular room measures 5 m by 4 m. Carpet costs £12 per m². What is the total cost of carpeting the room?

6. A square garden has an area of 121 m². What is the perimeter of the garden?

7. Select ALL of these statements that are TRUE about a parallelogram with base 15cm and height 4cm: A) its area is 60cm² B) its area is 30cm² C) it has the same area as a 15cm×4cm rectangle D) it has the same area as a triangle base 15cm height 8cm

8. A 30cm×20cm piece of card has a 4cm×4cm square cut from each of its four corners. What is the remaining area?

9. A 1.5 m wide path runs all around the outside of a 10m×6m garden. What is the total area, including the garden and the path?

10. A rectangle's length is 3 cm more than its width. If its perimeter is 34 cm, what are its dimensions?

11. A large rectangle 14m×9m has a 6m×4m rectangle removed from a corner, forming an L-shape. What is the perimeter of the L-shape?

12. A rectangular sports pitch is 100 m long and 64 m wide. A running track goes around the outside with a uniform width of 3 m. What is the perimeter of the outer edge of the track?

13. A parallelogram-shaped field has a base of 25 m and a height of 16 m. If 1 hectare = 10,000 m², what is the field's area in hectares?

14. A square and a rectangle have equal areas. The square has sides of 12 cm and the rectangle's width is 9 cm. What is the rectangle's length?

15. An L-shaped room is made of two rectangles, 6m×4m and 3m×2m (no overlap). Carpet costs £15 per m². What is the total cost to carpet

the room?